

Generative Deep Learning for Anatomy-Aware 3D Medical Image Synthesis

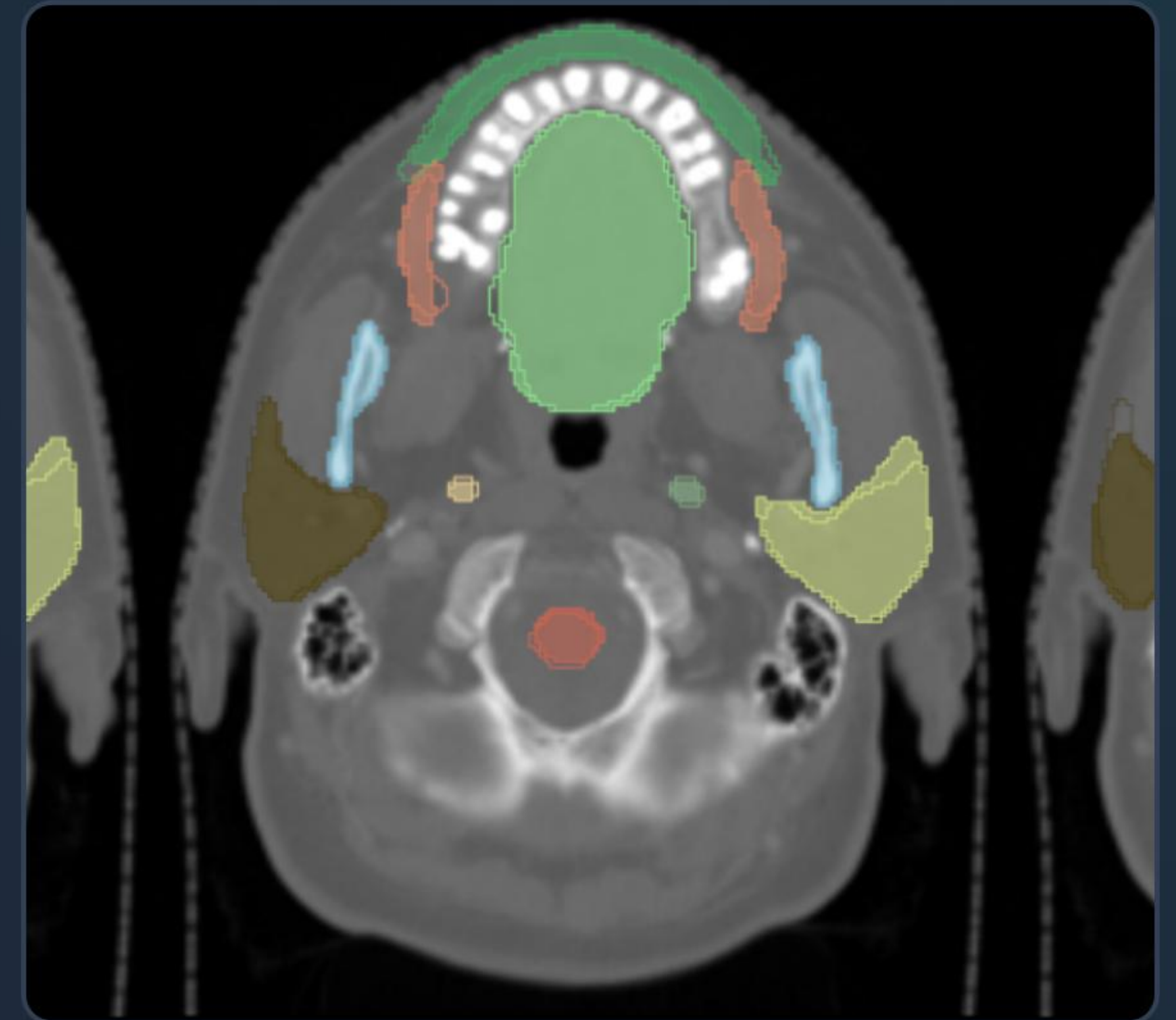
Text-to-CT Synthesis for Head and Neck Oncology

Under the guidance of : Dr. Amareswararao Kavuri

Submitted by : Atharva Rajesh Gosavi (252SP009)

Problem Statement

- ▶ **1. Context & Motivation:** Deep learning in the medical field is severely bottlenecked by data scarcity, strict privacy regulations, and the high cost of expert annotations.
- ▶ **2. The Spatial/Computational Barrier:** Generating full 3D CT volumes natively strains GPU memory beyond practical limits. Standard methods fail to scale efficiently.
- ▶ **3. The Domain Gap:** While chest CT synthesis exists, Head & Neck oncology—characterized by highly complex anatomy—entirely lacks publicly available text-paired datasets.
- ▶ **The Combined Bottleneck:** We lack a robust pipeline to generate diverse, patient-specific 3D anatomy conditioned purely on clinical text without requiring manual segmentation masks at inference.



Project Objectives



Primary Novelty

Design a **conditional Latent Diffusion Model (LDM)** capable of synthesizing highly accurate 3D Head & Neck CT volumes entirely from free-form clinical text (e.g., HPV status, TNM staging) using a frozen PubMedBERT encoder.



Pipeline Efficiency

Overcome severe 3D GPU memory constraints by integrating a pretrained **MAISI Variational Autoencoder (VAE)** to compress CTs into a latent space, and train a 3D Diffusion UNet from scratch with Classifier-Free Guidance.



Verification & Benchmarking

Implement an automated pseudo-report generation pipeline to overcome the lack of real reports. Evaluate generation fidelity using MAE, SSIM, PSNR, and HU Histogram Intersection against real HECKTOR 2026 scans.

References

1. Primary Baseline

Title: "MedSyn: Text-guided anatomy-aware synthesis of high-fidelity 3D CT images"

Journal / Year: IEEE Transactions on Medical Imaging (2024)

Relevance: Establishes the foundational pipeline (VAE + text encoder + diffusion UNet) for synthesizing chest CTs. Serves as the direct baseline that our work extends to the infinitely more complex Head and Neck domain.

2. Supporting Literature

a) Latent Compression Gap:

Title: "MAISI: Medical AI for synthetic imaging" (2024)

Relevance: Validates the VAE architecture used to compress full-resolution CTs into a viable latent space.

b) Domain-Specific Text Encoding:

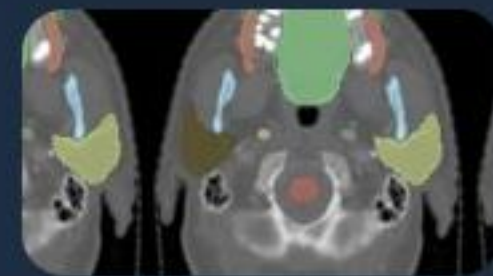
Title: "Domain-specific language model pretraining for biomedical natural language processing" (PubMedBERT)

Relevance: Proves the necessity of specialized encoders to understand clinical concepts like TNM staging over general models.

THANK YOU

Questions?

Image Sources



https://sandbox-cic-2021.sites.olt.ubc.ca/files/2024/07/11hanseg_prediction_vis.png

Source: cic.ubc.ca